

Introduction to Machine Learning

Machine Learning: Evolution & Impact

- Over 20 years ago, IBM's Deep Blue defeated world chess champion Garry Kasparov, bringing global attention to artificial intelligence and machine learning.
- Today, machine learning (ML) is a mature field, integrated into everyday life — from product and content recommendations to medical diagnostics and energy optimization.
- ML assists in diverse areas including:
 - Personalized recommendations (e.g., toys, books, literature)
 - Market predictions for traders
 - Medical decisions (tumor diagnosis)
 - Energy consumption optimization for environmental benefit

Machine Learning: Evolution & Impact

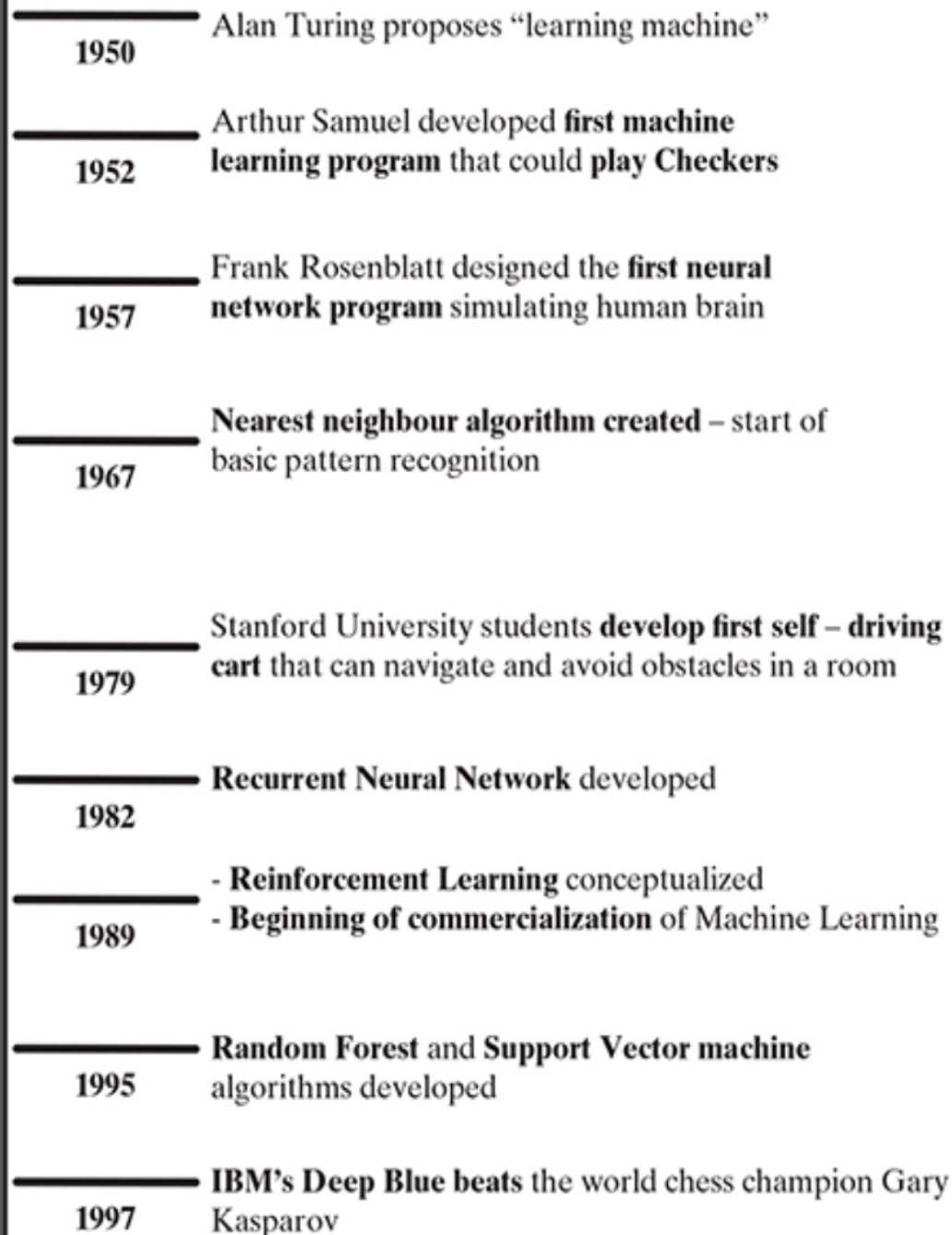
- Leading technology companies, especially Google, are at the forefront, with projects like Google Brain and self-driving cars fueled by ML innovation.
- The conceptual roots of machine learning date back to the 18th and 19th centuries:
 - 1763: Thomas Bayes's pioneering work (Bayes' Theorem)
 - 1805: Formalization of the least squares method (foundation for regression)
 - 1812: Bayes' theorem further developed by Pierre-Simon Laplace
 - 1913: Introduction of Markov chains

Machine Learning: Evolution & Impact

- Key milestones in modern machine learning:
 - 1950: Alan Turing questions if machines can think, laying AI's conceptual groundwork.
 - 1952: Arthur Samuel (IBM) developed early ML programs for Checkers.
 - 1957: Frank Rosenblatt created the first neural network simulating the human brain.
 - New algorithms emerged in the late 20th century:
 - 1969: Nearest neighbour algorithm
 - 1982: Recurrent neural network
 - 1995: Support vector machines, random forest algorithms
 - 2016: Google's AlphaGo beat a professional Go player, pushing the limits of ML in complex decision-making.

Machine Learning: Evolution & Impact

- ML has become deeply embedded in modern society and technological progress, impacting almost every aspect of daily life.
- The rapid development in the area of machine learning has triggered a question in everyone's mind – can machines learn better than human?
- To find its answer, the first step would be to understand what learning is from a human perspective.



A vertical timeline on the left side of the image, featuring a dark grey vertical bar on the far left. Horizontal lines extend from this bar to the right, each marking a year and a corresponding milestone in machine learning history. The milestones are listed in chronological order from top to bottom.

- 1950** Alan Turing proposes “learning machine”
- 1952** Arthur Samuel developed **first machine learning program** that could **play Checkers**
- 1957** Frank Rosenblatt designed the **first neural network program** simulating human brain
- 1967** **Nearest neighbour algorithm created** – start of basic pattern recognition
- 1979** Stanford University students **develop first self – driving cart** that can navigate and avoid obstacles in a room
- 1982** **Recurrent Neural Network** developed
- 1989**
 - **Reinforcement Learning** conceptualized
 - **Beginning of commercialization** of Machine Learning
- 1995** **Random Forest and Support Vector machine** algorithms developed
- 1997** **IBM’s Deep Blue beats** the world chess champion Gary Kasparov



- 2006**
 - **First machine learning competition** launched byNetflix
 - Geoffrey Hinton conceptualizes **Deep Learning**
- 2010** **Kaggle**, a website for machine learning competitions, **launched**
- 2011** **IBM’s Watson beats two human champions in Jeopardy**
- 2016** **Google’s AlphaGo program beats unhandicapped professional human player**

What is Human Learning?

- In cognitive science, learning is typically referred to as the process of gaining information through observation.
- Learning involves acquiring information through observation.
- Purpose of learning:
 - Necessary for performing a range of daily and complex activities.
 - Prior information about a task improves performance and efficiency.
- Continued learning leads to:
 - Better task execution with fewer mistakes (e.g., homework, rocket launches).
 - Enhanced problem-solving using past experiences.

What is Human Learning?

- Human learning typically occurs in three ways:
 - Expert-guided learning: Learning directly from an expert or teacher.
 - Knowledge-based learning: Building understanding indirectly from previously acquired expert knowledge.
 - Self-learning: Learning independently through trial and error, including unsuccessful attempts.
- Each learning method can be illustrated using real-life examples.

Learning under expert guidance



- Early Childhood:
 - Infants learn traits, facts, and language directly from parents/guardians.
 - Example: Naming body parts and colours as taught by caregivers.
- School Phase:
 - Children learn basics (alphabets, numbers), building complexity over time (sentences, mathematics, science) from teachers.
 - Teachers possess prior knowledge to guide students.
- Higher Education:
 - Students specialize in fields (engineering, medicine, etc.), learning advanced concepts from expert instructors.
- Professional Life:
 - Practical, hands-on skills are acquired with the help of mentors and experienced colleagues.
 - On-the-job training is guided by those with real-world experience.

Learning guided by knowledge gained from experts

- Indirect Learning: People often use knowledge gained earlier in different contexts to solve new problems or make decisions.
- An essential part of learning also happens with the knowledge which has been imparted by teacher or mentor at some point of time in some other form/context.
- Examples:
 - A baby groups items by colour using past knowledge, even without being directly taught to do so.
 - A child identifies an odd word out (verb among nouns) using labelling skills taught earlier by a teacher.
 - A professional targets customers in a campaign using old guidance about customer preferences from a mentor.
- In all these situations, there is no direct learning. It is some past information shared on some different context, which is used as a learning to make decisions.



Learning by self



- In many situations, humans are left to learn on their own.
- A classic example is a baby learning to walk through obstacles.
- He bumps on to obstacles and falls down multiple times till he learns that whenever there is an obstacle, he needs to cross over it.
- He faces the same challenge while learning to ride a cycle as a kid or drive a car as an adult.
- Not all things are taught by others.
- A lot of things need to be learnt only from mistakes made in the past.
- We tend to form a check list on things that we should do, and things that we should not do, based on our experiences.

What is Machine Learning?

- **‘A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.’** - Tom Mitchell, Professor of Machine Learning Department, School of Computer Science, Carnegie Mellon University.
- What this essentially means is that a machine can be considered to learn if it is able to gather experience by doing a certain task and improve its performance in doing the similar tasks in the future.
- When we talk about past experience, it means past data related to the task.
- This data is an input to the machine from some source.

What is Machine Learning?

- Concept Mapping in ML:
 - *E (Experience)*: Past data or practice (e.g., checkers games played / labeled images).
 - *T (Task)*: The actual objective (e.g., playing checkers / classifying images).
 - *P (Performance)*: Metric to evaluate success (e.g., % of games won / % of correct classifications).
- Examples:
 - Checkers:
 - E = playing many games
 - T = playing checkers
 - P = % of games won
 - Image Classification:
 - E = labeled training images
 - T = classify unlabeled images
 - P = % of correct classifications

The first step in any project is defining your problem. Even if the most powerful algorithm is used, the results will be meaningless if the wrong problem is solved.

How do machines learn?

- The basic machine learning process can be divided into three parts.
- **Data Input:** Past data or information is utilized as a basis for future decision-making
- **Abstraction:** The input data is represented in a broader way through the underlying algorithm.
- **Generalization:** The abstracted representation is generalized to form a framework for making decisions



How do machines learn? (Cont..)

- **Example:** Learning from Classroom, books and preparing for the examination
- Many students try to memorize as many things as possible.
 - work well when the scope of learning is not so vast
 - questions are simple and straightforward
 - answered by simply writing the same things which have been memorized.
- If the questions get more complex, **a direct reproduction of the things memorized may not help.** the things memorized may not help.
- Human learning is that just by great memorizing and perfect recall, i.e. just based on knowledge input, students can do well in the examinations only till a certain stage.
- A good option for complex questions is:
 - figure out the key points or ideas amongst a vast pool of knowledge.
 - create an outline of topics
 - conceptual mapping of those outlined topics with the entire knowledge pool.

How do machines learn? (Cont..)

- For example, a broad pool of knowledge may consist of all living animals and their characteristics such as
 - whether they live in land or water,
 - whether they lay eggs
 - whether they have scales or fur or none ,etc.
- It is a difficult task for any student to memorize the characteristics of all living animals.
- It is better to draw a notion about the basic groups that all living animals belong to and the characteristics which define each of the basic groups.
 - The basic groups of animals are invertebrates and vertebrates.
 - Vertebrates are further grouped as mammals, reptiles, amphibians, fishes and birds.
- Here, we have mapped animal groups and their salient characteristics.

How do machines learn? (Cont..)

- Moving to the machine learning paradigm, the **vast pool of knowledge is available from the data input.**
- A **concept map to derive the characteristics is drawn from the input data. This is nothing but knowledge abstraction** as performed by the machine.
- In the end, the abstracted mapping from the input data can be applied to make critical conclusions.
- For example, if the group of an animal is given, understanding of the characteristics can be automatically made.
- Reversely, if the characteristic of an unknown animal is given, a definite conclusion can be made about the animal group it belongs to.
- This is **generalization** in context of machine learning.

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Abstraction

- In machine learning, the process always begins with raw input data.
- However, this data is often too complex, noisy, or unstructured to be used directly.
- This is where abstraction comes into play.
- Abstraction means simplifying the data into a representation that captures only the essential knowledge.
- The result of this process is a conceptual map or a model, which becomes the foundation for making predictions, drawing insights, or identifying patterns.

Forms of Abstraction (Types of Models)

- Models, which are abstractions of raw data, can take different forms depending on the situation:
 - Rule-based models (e.g., if/else conditions) are useful for decision logic.
 - Mathematical equations such as linear or logistic regression represent relationships numerically.
 - Data structures like trees or graphs organize observations for classification or network analysis.
 - Logical groupings such as clusters help identify similarities within datasets.
- Each form is essentially a different way to abstract knowledge from the raw input.

Choosing the Right Abstraction (Model)

- The model you choose is not automatic—it depends on different aspects:
 - **Type of problem:** For forecasting or prediction, a regression model might be better, whereas for grouping similar objects, a clustering model fits.
 - **Nature of input data:** Models must deal with issues like missing values, different types of data, or the sheer size of datasets.
 - **Domain requirements:** In critical domains like banking, models need to give immediate and reliable results (e.g., fraud detection must happen in real time).
- So, abstraction is always a human-guided decision to suit the problem, the data, and the domain.

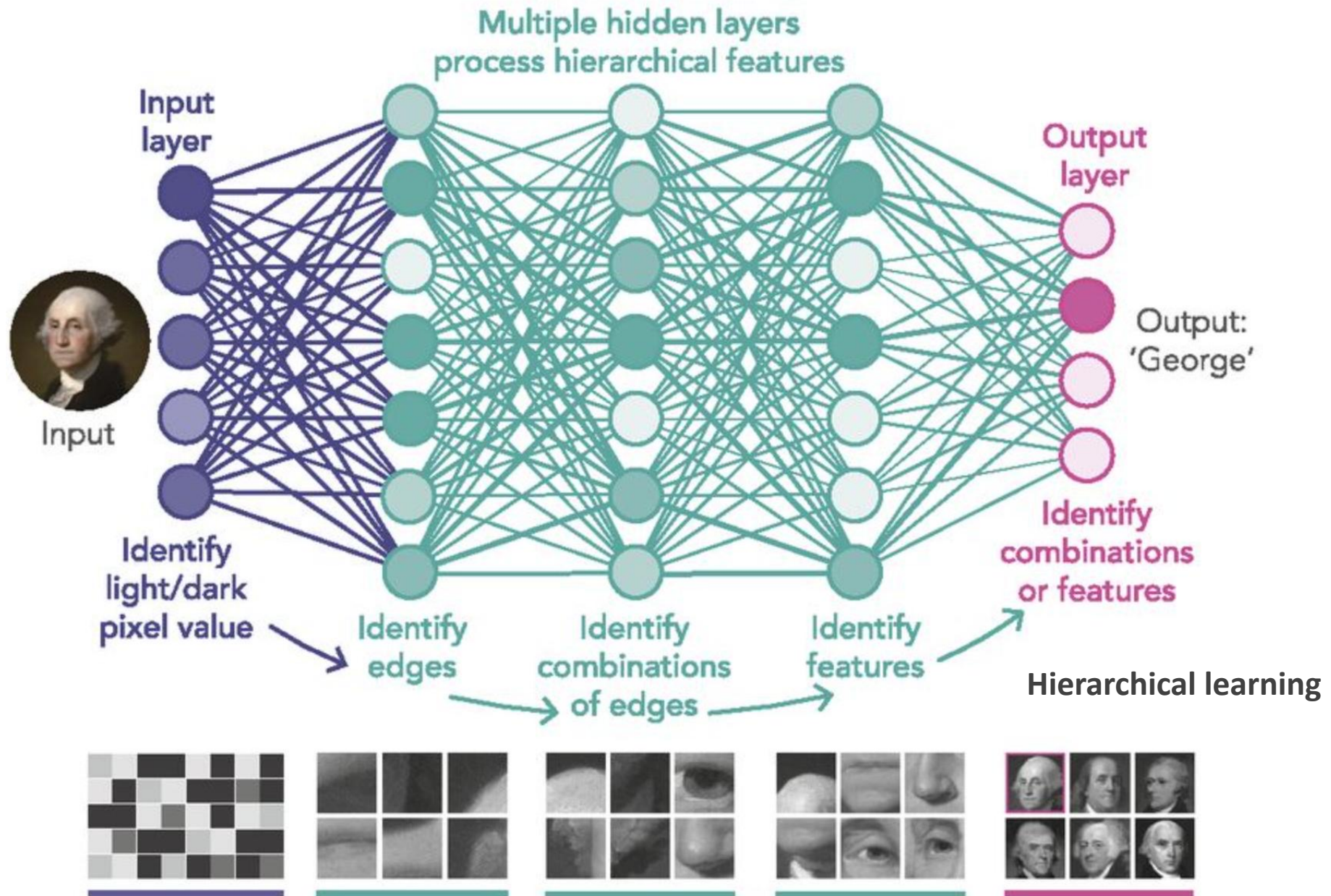
Abstraction in Action: Example

- Let's consider simple linear regression as an example of abstraction.
- The model is represented by the equation $y = c_1 + c_2x$.
- This is just an abstract mathematical form until we determine what the constants c_1 and c_2 should be.
- Using the input data, the process of finding these values is called fitting the model.
- When we do this, the abstract equation becomes meaningful because it represents the real relationship between x (input) and y (output).
- This process of fitting the abstraction to data is called training, and the dataset used is known as training data.

Abstraction

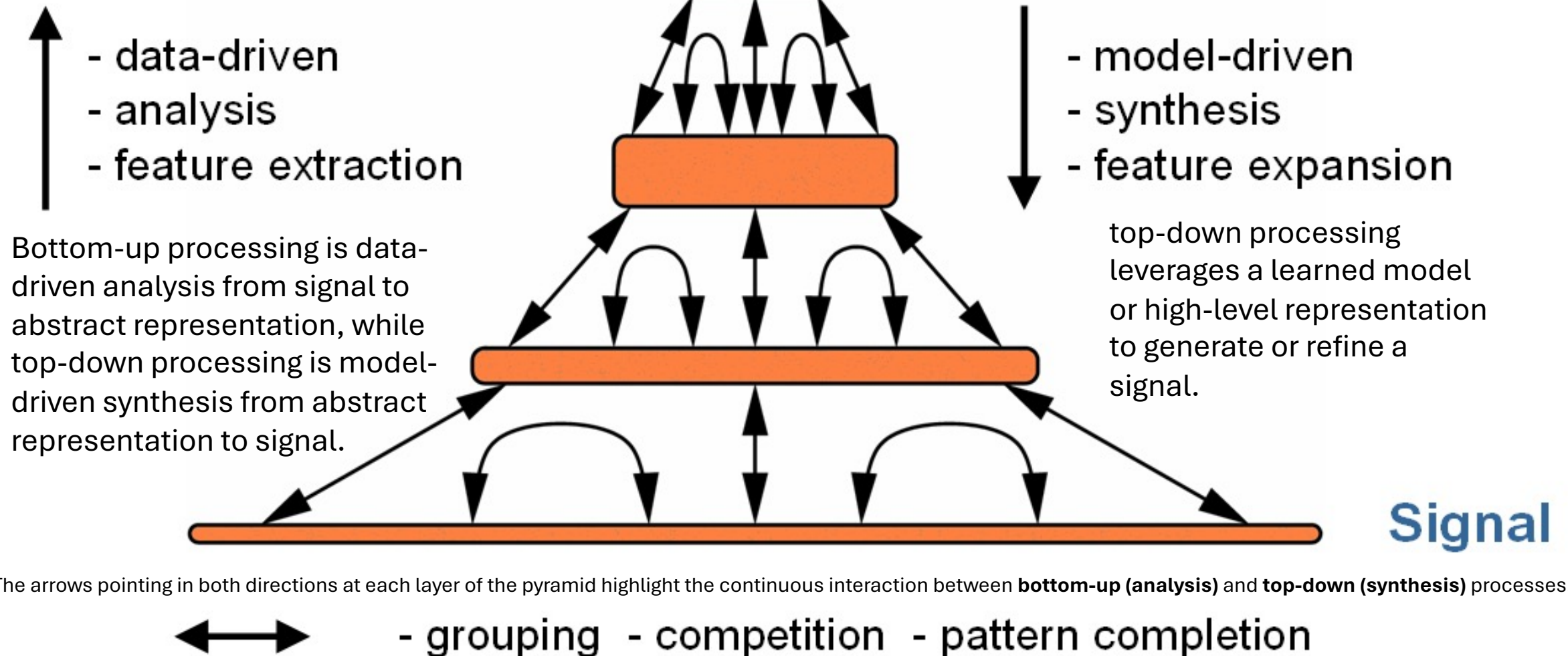
- To summarize, abstraction in machine learning is the process of simplifying raw data into a meaningful structure called a model.
- This model could take the form of rules, equations, data structures, or logical groupings.
- Selecting the right level and type of abstraction depends on the problem, the nature of input data, and the domain in which the model will be applied.
- Finally, once a model is chosen, it must be trained with data to make it usable.
- Abstraction transforms unorganized data into knowledge that can drive predictions, analysis, and decision-making.

DEEP LEARNING NEURAL NETWORK

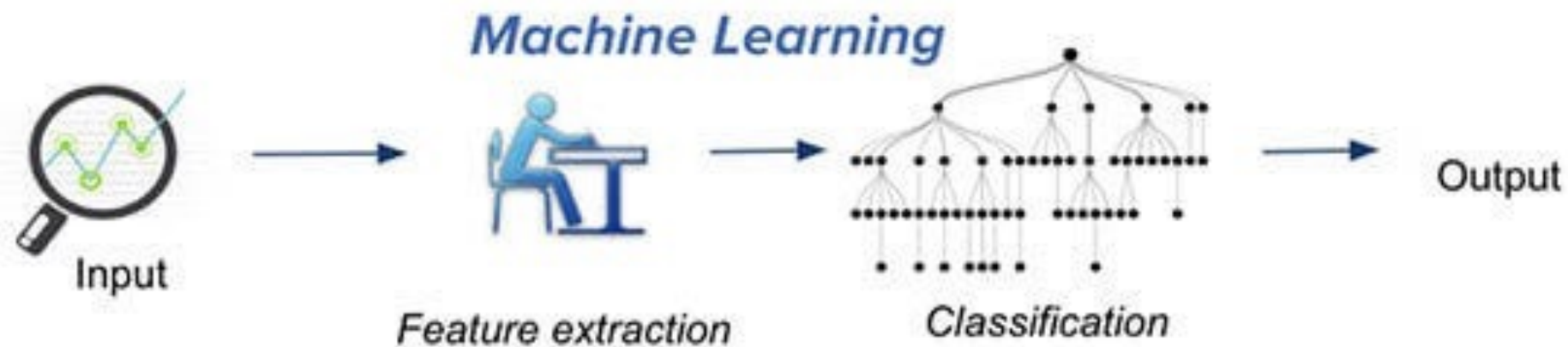


Neural Abstraction Pyramid

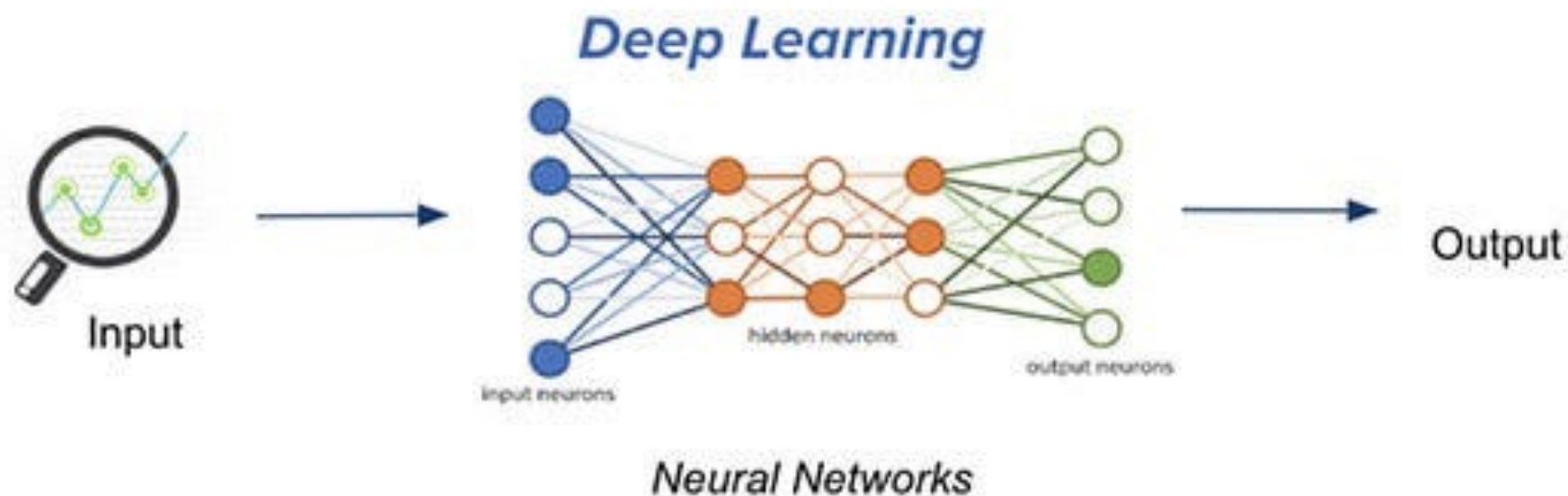
[S. Behnke: Hierarchical Neural Networks for Image Interpretation, Springer 2003]



The arrows pointing in both directions at each layer of the pyramid highlight the continuous interaction between **bottom-up (analysis)** and **top-down (synthesis)** processes.



Traditional machine learning uses hand-crafted features, which is tedious and costly to develop.



Deep learning learns hierarchical representation from the data itself, and scales with more data.

Generalization

- The first stage of machine learning involves abstraction, where input data is transformed into a model.
- This step is commonly referred to as training the model.
- Training allows the system to capture patterns and relationships present in the data.
- However, abstraction alone does not make the model useful for real-world decision-making.
- The ultimate goal is to adjust and refine this knowledge so that it can be applied effectively to predict outcomes for new, unseen situations.
- This critical refinement process is called generalization, and achieving it is both necessary and challenging.

Generalization

- Generalization focuses on enabling a trained model to make accurate predictions when faced with unfamiliar test data.
- Since training is performed on a limited dataset, the model only captures part of reality.
- Two challenges usually occur: first, the risk of overfitting, where the model becomes too specifically tied to the training data;
- second, the possibility that test data may contain patterns absent from the training set.
- These challenges reduce the reliability of precise decision-making approaches.
- As a result, an approximate or flexible strategy is required when applying the model to new data.

Generalization

- To handle these limitations, machine learning models often rely on heuristic or approximate approaches, rather than precise, fixed rules.
- This is similar to how human beings depend on intuition or gut feelings when exact reasoning is not possible.
- Such methods allow models to work successfully on unseen inputs, but there is always a risk of incorrect outcomes because assumptions made during training may not hold in reality.
- Even so, this uncertainty is common to both machines and humans, highlighting the essential balance between accuracy, generalization, and the acceptance of potential errors in decision-making.

Well-posed learning problem

- For defining a new problem, which can be solved using machine learning, a simple framework, highlighted below, can be used.
- This framework also helps in deciding whether the problem is a right candidate to be solved using machine learning. The framework involves answering three questions:
 1. What is the problem?
 2. Why does the problem need to be solved?
 3. How to solve the problem?

Step 1: What is the Problem?

- The first step in framing a machine learning problem is defining what the problem actually is.
- Start with an informal description.
- For example: “*I need a program that can predict the next word as I type.*” Then, formalize the definition using Tom Mitchell’s model of Task (T), Experience (E), and Performance (P).
- For instance:
 - Task (T): Suggest the next word.
 - Experience (E): A corpus of common English words.
 - Performance (P): Accuracy of suggested words.
- Additionally, list assumptions and identify similar existing problems. What other problems have you seen or can you think of that are similar to the problem that you are trying to solve?

Step 2: Why Solve the Problem?

- Once the problem is defined, the next step is clarifying why it needs to be solved.
- This involves identifying the motivation behind the problem and the requirements it fulfils.
- For example, does it help detect fraudulent transactions (critical business advantage), or is it aimed at providing convenience, like recommending movies?
- Clearly articulating the benefits is key—whether it enables cost savings, efficiency improvements, or personalization.
- These benefits should also be positioned as justification for project approval.
- Finally, consider how the solution will be used and the expected lifetime of its value.

Step 3: How to Solve the Problem?

- After identifying the “what” and “why,” the next stage is “how.” Begin by envisioning how you would solve the problem manually.
- Lay out detailed steps for data collection, preparation, and program design.
- This process helps refine assumptions made earlier and creates a plan for implementation.
- For instance, solving a predictive typing task would involve collecting text data, cleaning the corpus, designing algorithms, and finally, training the model.
- These documented steps not only help create clarity in design but also ensure effective alignment between problem definition and technical approach.

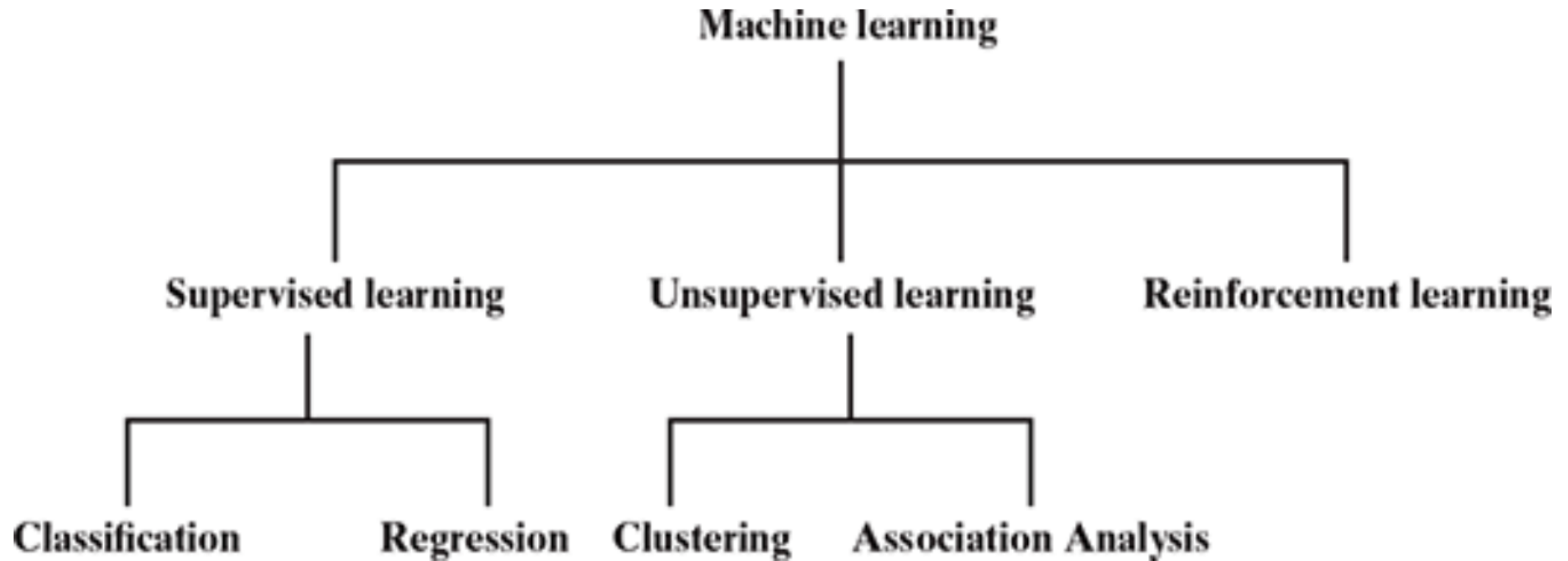
All 3 Steps...

- Step 1: What is the problem? Describe the problem informally and formally and list assumptions and similar problems.
- Step 2: Why does the problem need to be solved? List the motivation for solving the problem, the benefits that the solution will provide and how the solution will be used.
- Step 3: How would I solve the problem? Describe how the problem would be solved manually to flush domain knowledge.

Types of Machine Learning

- Machine learning can be classified into three broad categories:
 1. Supervised learning – Also called predictive learning. A machine predicts the class of unknown objects based on prior class-related information of similar objects.
 2. Unsupervised learning – Also called descriptive learning. A machine finds patterns in unknown objects by grouping similar objects together.
 3. Reinforcement learning – A machine learns to act on its own to achieve the given goals.

Types of Machine Learning



Supervised learning

- The main motivation of supervised learning is to enable machines to learn from past information, much like humans do.
- This past information is what we call ‘experience.’ With this experience, machines become capable of solving tasks without being explicitly programmed for every situation.
- The key idea is that the more relevant past information we provide, the better the machine becomes at making accurate predictions in the future.
- In essence, supervised learning uses previous data to guide the system, allowing it to handle new, unseen inputs effectively and with improved accuracy.

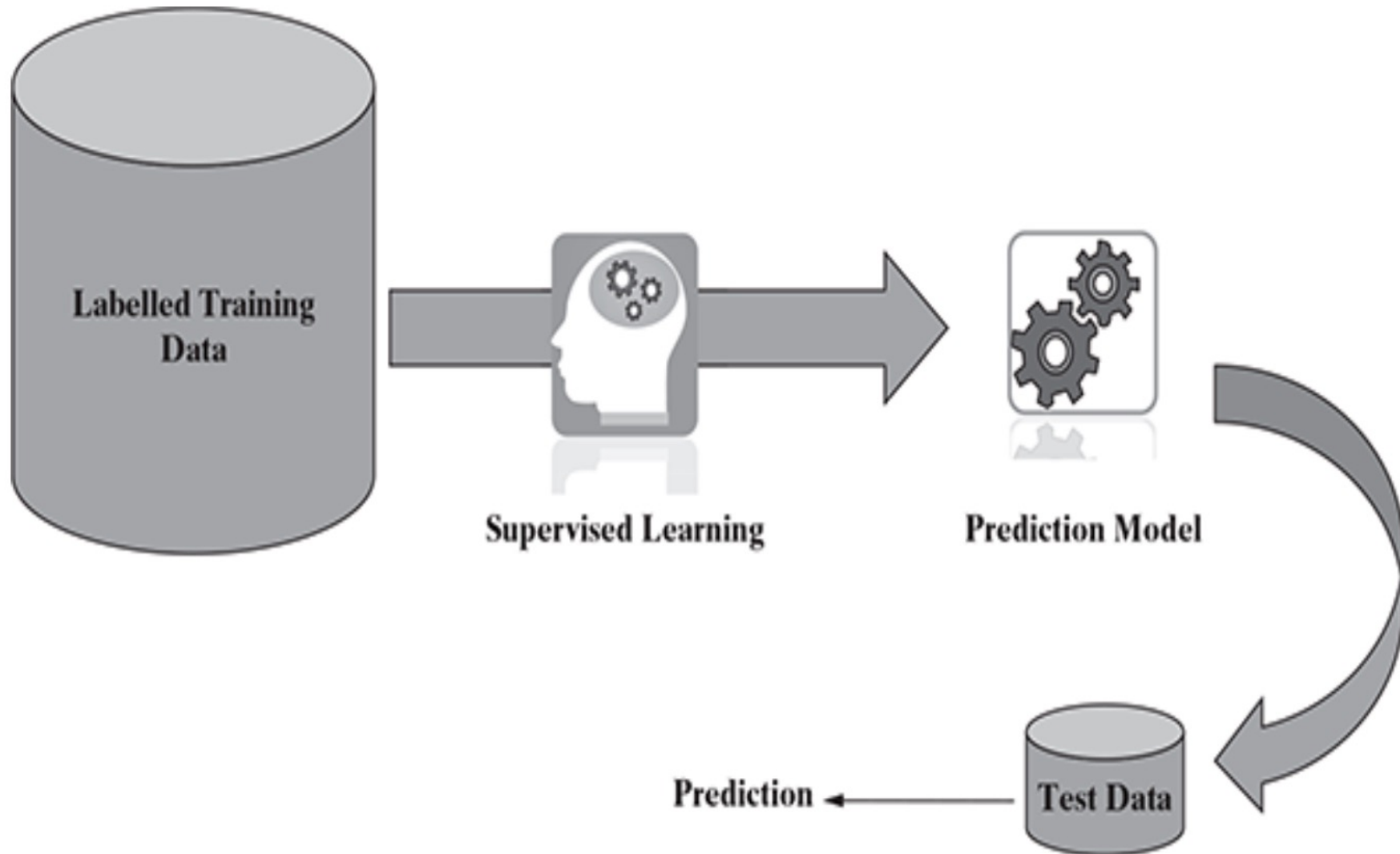
Supervised learning

- For supervised learning to work, the machine needs past information specifically related to the task at hand.
- If the task is image identification, the past information must describe features of shapes or colours.
- In machine learning terms, this past information is called *experience*.
- Experience does not mean real-world general knowledge, but well-structured data collected from previous examples of the task.
- A supervised learning model relies on this experience to find patterns, understand differences between categories, and learn how to separate or classify new examples successfully.
- This past data is the model's foundation.

Supervised learning

- The essential input that enables supervised learning is called *training data*.
- Training data represents past experiences relevant to a particular task.
- In the case of image classification, training data contains various images with details about features, such as shapes or colours. Crucially, each example in the training data includes a tag known as a *label*.
- A label specifies the correct category for that example, such as “round” or “blue.” This labelling is critical, because it tells the machine the ground truth while learning.
- Supervised learning specifically depends on such labelled training data.

Supervised learning



Examples of Supervised Learning

- Supervised learning is widely applied in practical, real-world scenarios.
- Some common examples include: predicting the results of a sports game, predicting whether a medical tumour is malignant or benign, and predicting financial aspects such as the price of real estate or stocks.
- It is also used in classifying texts, such as identifying emails as spam or non-spam.
- Each of these cases involves using past labelled data to train the machine.
- This allows the system to make informed decisions or predictions when it encounters new, unseen inputs that are similar in nature.

Comparing Two Example Problems

- Let's focus on two problems: predicting whether a tumour is malignant or benign, and predicting the price of real estate.
- Though both are prediction tasks, they are not the same in nature. Tumour prediction involves determining whether a patient falls into one of two categories—malignant or benign.
- Real estate prediction, on the other hand, deals with predicting an exact value such as the price of a house.
- This shows that supervised learning problems, though similar in concept, can differ based on whether the outcome is categorical or numerical.

Classification Problems

- When the task involves predicting a categorical or nominal variable, it falls under classification.
- For example, in cancer diagnosis, the system predicts whether a tumour belongs to the malignant class or the benign class.
- Similarly, spam detection requires classifying an email as either spam or not spam.
- Classification always results in assigning input data to one of several predefined categories.
- This type of supervised learning is crucial in domains like healthcare, fraud detection, and text analysis, where outcomes are distinct labels.
- The focus is on accurate separation of data into clear, fixed classes.

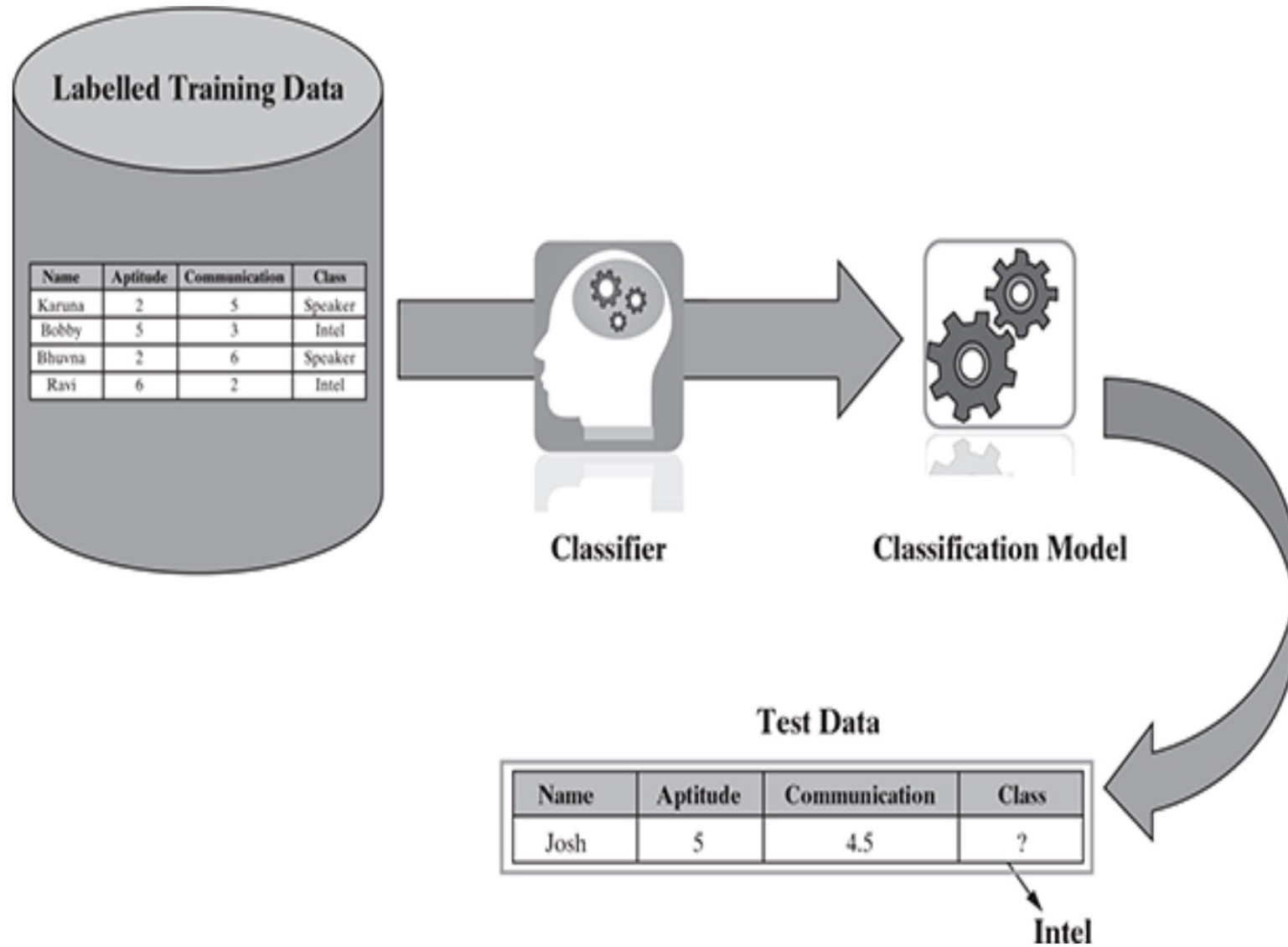
Regression Problems

- When the task involves predicting a continuous, real-valued variable, it is known as regression. For instance, predicting real estate prices or stock values involves estimating actual numbers rather than assigning categories.
- Unlike classification, regression problems focus on finding patterns that allow for forecasting values along a continuous scale.
- The machine uses past labelled data with numerical outcomes to build models that can estimate future results.
- Regression is especially useful in finance, economics, and business forecasting, where precise numerical predictions are essential for effective planning and decision-making.

Classification

- To understand classification, let's consider the task of segregating images based on shapes.
- For example, if an image contains a round object, it is placed into one category, while a triangular object is placed into another.
- The challenge arises when we encounter a new, unseen image, also known as *test data*.
- The machine decides its category by comparing the test image to past labelled data, called *training data*.
- Since each training example already has a category defined, the model assigns the same label to new test data by measuring similarity.

Classification



Classification

- The essence of a classification problem is assigning a *label, category, or class* to new test data based on the labelled examples from past training data.
- In supervised learning, this means the model makes predictions about which class an unseen input belongs to.
- The objective is not to predict a number but to classify the data into predefined buckets.
- Classification therefore plays a key role in a variety of fields where the decision involves labelling inputs correctly.
- This is one of the most common forms of supervised learning and serves as its foundational concept.

Classification

- There are several widely used machine learning algorithms that help solve classification problems efficiently. Some of the most popular include:
- Naïve Bayes: Based on probability and Bayes' theorem.
- Decision Trees: Use logical rules to split data into branches.
- k-Nearest Neighbour (k-NN): Classifies data based on similarity to its closest neighbours.
- These algorithms have been successfully applied in different industries, ranging from healthcare and banking to text recognition. The choice of algorithm depends on the type of data, required accuracy, and computational efficiency of the task.

Classification in Banking – Fraud Detection

- A critical real-world application of classification is fraud detection in the banking sector.
- With millions of transactions happening daily, it is impossible for humans to manually inspect all of them.
- Machine learning helps automate this task by using past labelled transaction data where fraudulent cases are already identified.
- The system learns patterns associated with fraud and then labels new transactions as either normal or suspicious.
- Suspicious transactions are flagged for further review by experts.
- This is a classic example of classification in action, protecting both banks and customers from financial risks and fraudulent activities.

Typical Examples of Classification Problems

- Classification problems exist across numerous domains. Some typical examples include:
 - Image Classification – identifying objects in images.
 - Disease Prediction – diagnosing illnesses from patient data.
 - Win–Loss Predictions – forecasting game or sports outcomes.
 - Natural Calamity Prediction – identifying risks of earthquakes or floods.
 - Handwriting Recognition – converting handwritten text into a digital form.
- All these tasks involve mapping input data into categories or labels. These examples highlight the versatility of classification as one of the most powerful tools for supervised learning applications.

Regression

- In linear regression, the objective is to predict numerical features like real estate or stock price, temperature, marks in an examination, sales revenue, etc.
- The predictor variable and the target variable are continuous in nature.
- In linear regression, a straight line relationship is fitted between the predictor variables and the target variables using the least squares method.
- The least squares method works by minimizing the sum of squared errors between actual and predicted values of the target variable.

Regression

- Simple Linear Regression: model with only one predictor variable.
- Multiple Linear Regression: model with multiple predictor variables.
- Example (Sales Managers' Budgeting): They predict sales for the next year.
- Prediction is based on past sales figures and level of investment.
- Both past data and data to be predicted are continuous in nature.
- In a simple approach, investment = predictor variable, sales revenue = target variable.

Regression

- A typical simple regression model, where regression line is fitted based on values of target variable with respect to different values of predictor variable.
- A typical linear regression model can be represented in the form

$$y = \alpha + \beta x$$

- where 'x' is the predictor variable and 'y' is the target variable.

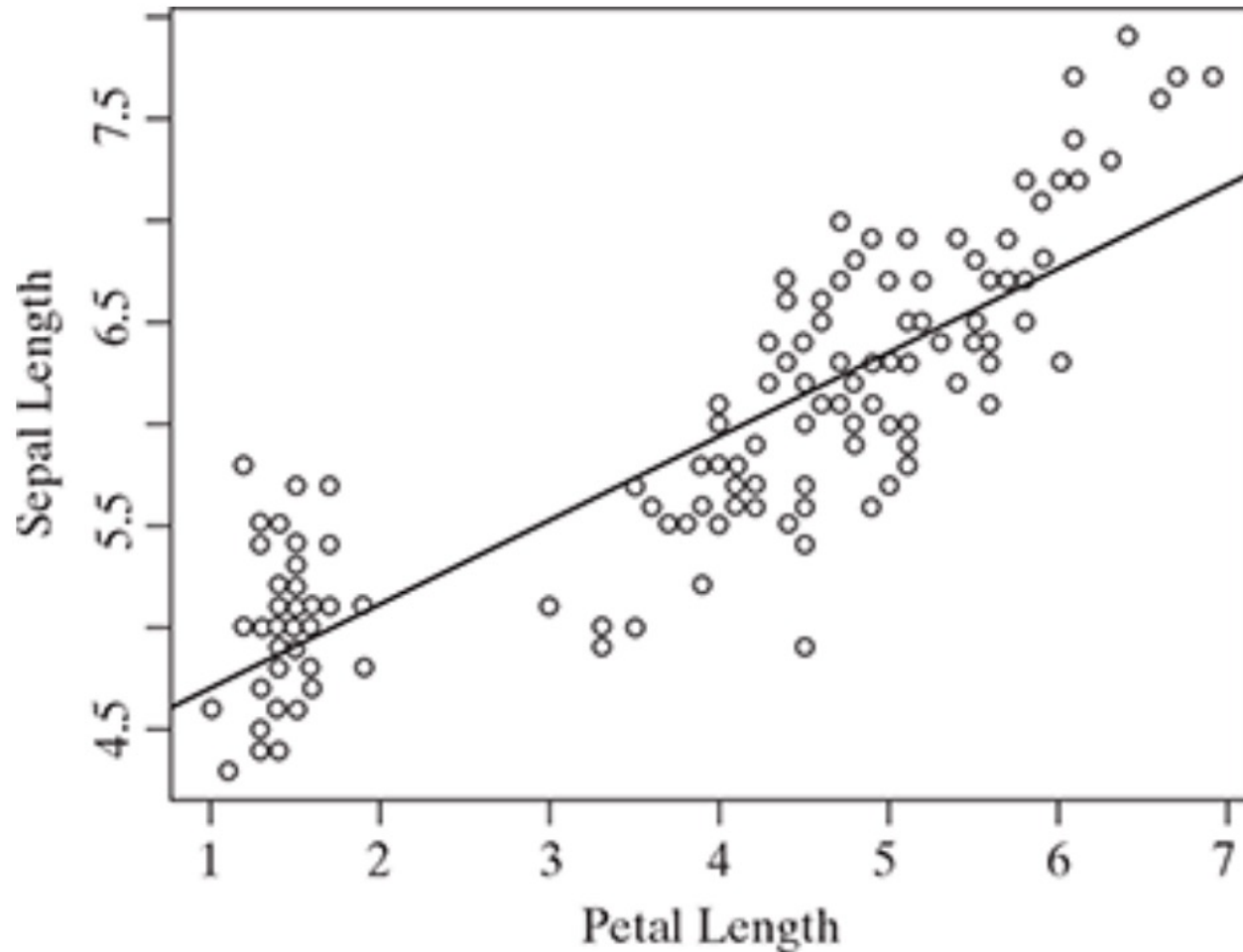
Regression

- Lets look at a regression model for input data that comes from a famous multivariate data set named Iris introduced by the British statistician and biologist Ronald Fisher.
- The data set consists of 50 samples from each of three species of Iris – Iris setosa, Iris virginica, and Iris versicolor.
- Four features were measured for each sample: sepal length, sepal width, petal length, and petal width.
- These features can uniquely discriminate the different species of the flower.

Regression

- The Iris data set is typically used as a training data for solving the classification problem of predicting the flower species based on feature values.
- However, we can also demonstrate regression using this data set, by predicting the value of one feature using another feature as predictor.
- Here petal length is a predictor variable which, when fitted in the simple linear regression model, helps in predicting the value of the target variable sepal length.

Regression



Typical applications of regression can be seen in

- **Demand forecasting in retails**
- **Sales prediction for managers**
- **Price prediction in real estate**
- **Weather forecast**
- **Skill demand forecast in job market**

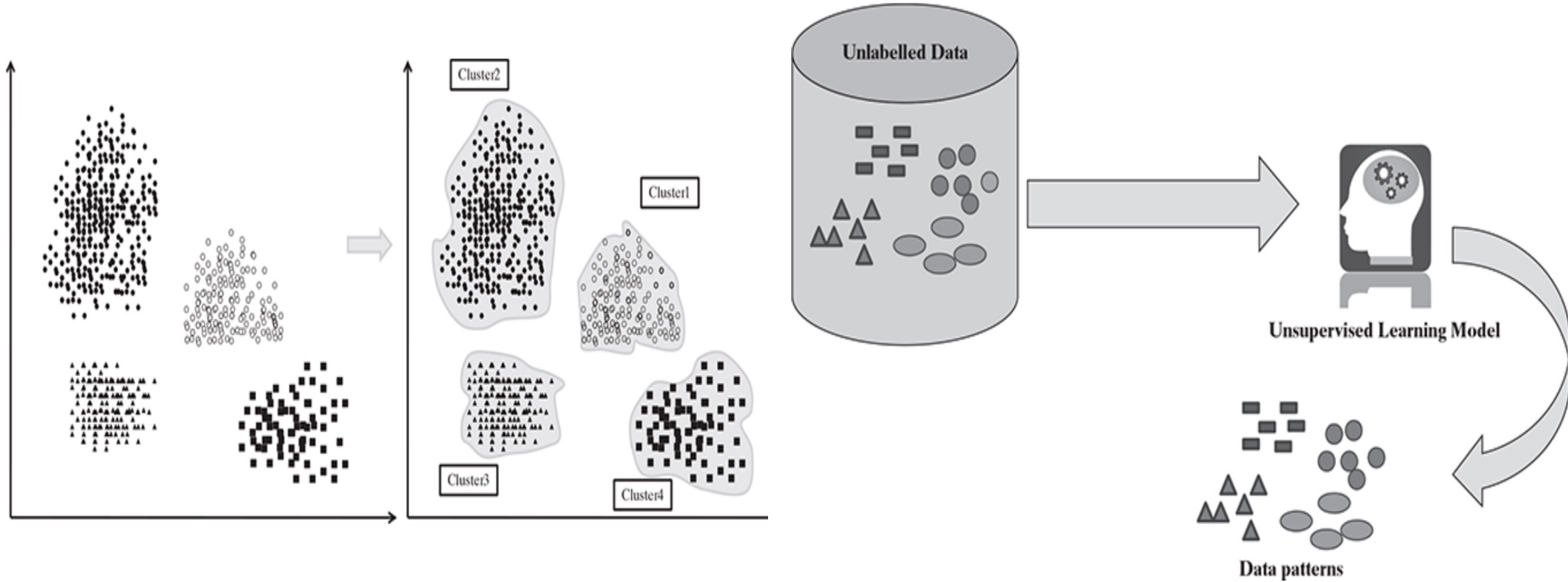
Unsupervised learning

- Unlike supervised learning, in unsupervised learning, there is no labelled training data to learn from and no prediction to be made.
- In unsupervised learning, the objective is to take a dataset as input and try to find natural groupings or patterns within the data elements or records.
- Therefore, unsupervised learning is often termed as descriptive model and the process of unsupervised learning is referred as pattern discovery or knowledge discovery.
- One critical application of unsupervised learning is customer segmentation.

Unsupervised learning

- Clustering is the main type of unsupervised learning.
- It intends to group or organize similar objects together.
- For that reason, objects belonging to the same cluster are quite similar to each other while objects belonging to different clusters are quite dissimilar.
- Hence, the objective of clustering is to discover the intrinsic grouping of unlabelled data and form clusters.
- Different measures of similarity can be applied for clustering.
- One of the most commonly adopted similarity measure is distance.
- Two data items are considered as a part of the same cluster if the distance between them is less.
- In the same way, if the distance between the data items is high, the items do not generally belong to the same cluster. This is also known as distance-based clustering.

Unsupervised learning



Unsupervised Learning

- Other than clustering of data and getting a summarized view from it, one more variant of unsupervised learning is association analysis.
- As a part of association analysis, the association between data elements is identified.
- Let's try to understand the approach of association analysis in context of one of the most common examples, i.e. market basket analysis

Unsupervised Learning

- From past transaction data in a grocery store, it may be observed that most of the customers who have bought item A, have also bought item B and item C or at least one of them.
- This means that there is a strong association of the event 'purchase of item A' with the event 'purchase of item B', or 'purchase of item C'.
- Identifying these sorts of associations is the goal of association analysis.
- This helps in boosting up sales pipeline, hence a critical input for the sales group.
- Critical applications of association analysis include market basket analysis and recommender systems.

Unsupervised Learning

TransID	Items Bought
1	{Butter, Bread}
2	{Diaper, Bread, Milk, Beer}
3	{Milk, Chicken, Beer, Diaper}
4	{Bread, Diaper, Chicken, Beer}
5	{Diaper, Beer, Cookies, Ice cream}
...	...

Market Basket transactions

Frequent itemsets → (Diaper, Beer)

Possible association: Diaper → Beer

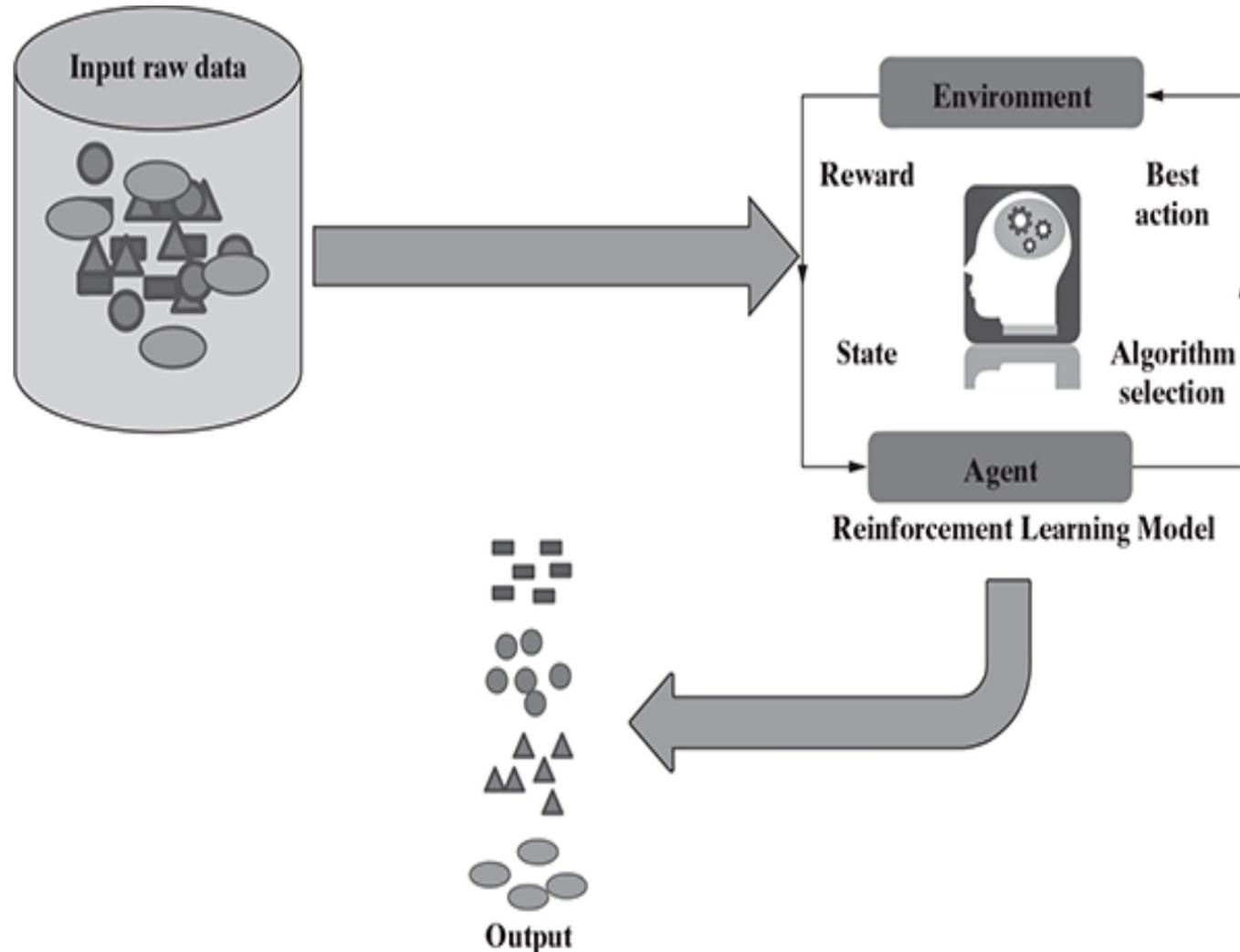
Reinforcement learning

- We have seen babies learn to walk without any prior knowledge of how to do it. Often we wonder how they really do it.
- They do it in a relatively simple way.
- First they notice somebody else walking around, for example parents or anyone living around. They understand that legs have to be used, one at a time, to take a step.
- While walking, sometimes they fall down hitting an obstacle, whereas other times they are able to walk smoothly avoiding bumpy obstacles.
- When they are able to walk overcoming the obstacle, their parents are elated and appreciate the baby with loud claps / or may be a chocolates.
- When they fall down while circumventing an obstacle, obviously their parents do not give claps or chocolates.
- Slowly a time comes when the babies learn from mistakes and are able to walk with much ease.

Reinforcement learning

- In the same way, machines often learn to do tasks autonomously.
- Let's try to understand in context of the example of the child learning to walk.
- The action tried to be achieved is walking, the child is the agent and the place with hurdles on which the child is trying to walk resembles the environment.
- It tries to improve its performance of doing the task.
- When a sub-task is accomplished successfully, a reward is given.
- When a sub-task is not executed correctly, obviously no reward is given.
- This continues till the machine is able to complete execution of the whole task.
- This process of learning is known as reinforcement learning.

Reinforcement learning



One contemporary example of reinforcement learning is self-driving cars.

The critical information which it needs to take care of are speed and speed limit in different road segments, traffic conditions, road conditions, weather conditions, etc.

The tasks that have to be taken care of are start/stop, accelerate/decelerate, turn to left / right, etc.

SUPERVISED	UNSUPERVISED	REINFORCEMENT
This type of learning is used when you know how to classify a given data, or in other words classes or labels are available.	This type of learning is used when there is no idea about the class or label of a particular data. The model has to find pattern in the data.	This type of learning is used when there is no idea about the class or label of a particular data. The model has to do the classification – it will get rewarded if the classification is correct, else get punished.
Labelled training data is needed. Model is built based on training data.	Any unknown and unlabelled data set is given to the model as input and records are grouped.	The model learns and updates itself through reward/ punishment.
The model performance can be evaluated based on how many misclassifications have been done based on a comparison between predicted and actual values.	Difficult to measure whether the model did something useful or interesting. Homogeneity of records grouped together is the only measure.	Model is evaluated by means of the reward function after it had some time to learn.
There are two types of supervised learning problems – classification and regression.	There are two types of unsupervised learning problems – clustering and association.	No such types.

SUPERVISED	UNSUPERVISED	REINFORCEMENT
Simplest one to understand.	More difficult to understand and implement than supervised learning.	Most complex to understand and apply.
Standard algorithms include • Naïve Bayes • <i>k</i>-nearest neighbour (kNN) • Decision tree • Linear regression • Logistic regression • Support Vector Machine (SVM), etc.	Standard algorithms are • <i>k</i>-means • Principal Component Analysis (PCA) • Self-organizing map (SOM) • Apriori algorithm • DBSCAN etc.	Standard algorithms are • Q-learning • Sarsa
Practical applications include • Handwriting recognition • Stock market prediction • Disease prediction • Fraud detection, etc.	Practical applications include • Market basket analysis • Recommender systems • Customer segmentation, etc.	Practical applications include • Self-driving cars • Intelligent robots • AlphaGo Zero (the latest version of DeepMind's AI system playing Go)

Applications of Machine Learning

Banking and Finance

- Machine learning plays a critical role in managing large-scale, high-velocity financial transactions.
- Fraudulent activities, especially in credit card transactions, are detected and prevented in real time, reducing disputes and operational costs. It is also applied to minimize customer churn.
- Descriptive learning identifies problem areas such as specific banks, branches, or product lines where churn is high.
- Predictive learning helps detect vulnerable customers who are likely to leave soon.
- Proactive measures, like tailored retention strategies, can then be implemented.
- Thus, ML strengthens fraud detection, ensures smoother operations, and helps banks retain valuable customers in a very competitive market.

Applications of Machine Learning

Insurance

- The insurance sector extensively leverages machine learning because of its data-intensive nature.
- Two major applications include risk prediction and claims management.
- During customer onboarding, ML models evaluate a prospective customer's risk profile based on historical information, which then drives premium quotations.
- In claims processing, ML helps detect fraud by analyzing past claims, adjuster notes, and customer history.
- Additionally, demographic and geographic data of similar customers are considered for accurate predictions.
- These techniques not only improve operational efficiency but also reduce financial losses.
- With predictive insights, the insurance industry can manage risks more effectively and ensure fair and timely settlements.

Applications of Machine Learning

Healthcare

- Machine learning enhances healthcare by analyzing data from wearable devices, enabling real-time health monitoring and early intervention.
- For example, wearable devices track vital signs such as blood pressure and transmit the data to remote servers where ML algorithms continuously analyze patterns.
- Using patient history and data of similar age groups, ML models can predict potential health threats.
- If an abnormal condition is detected—such as sudden spikes in blood pressure—alerts are sent to the individual and nearby healthcare providers.
- This allows immediate preventive action and rapid medical response, making healthcare more proactive and personalized, particularly for elderly patients.

Tools and Languages in Machine Learning

Python and R

- Python is the most popular open-source language for machine learning, first released in 1991 and managed by the Python Software Foundation.
- It has strong libraries like NumPy, SciPy, matplotlib, and scikit-learn for regression, classification, and clustering tasks.
- R, on the other hand, is a language mainly used for statistical computing and data analysis, widely popular in academia and among statisticians.
- With libraries like dplyr, caret, tm, and ggplot2, R is powerful for machine learning pipelines.
- Its packages such as Shiny and R Markdown also allow interactive dashboards and reporting, making it suitable for research and deployment.

Tools and Languages in Machine Learning

MATLAB and SAS

- MATLAB is a commercial licensed software developed by MathWorks, supporting a wide range of numerical and statistical computing tasks.
- It provides built-in algorithms for machine learning, strong documentation, and capabilities like parallel processing on clusters and cloud for scaling large datasets.
- SAS, earlier known as Statistical Analysis System, is another licensed commercial tool created by SAS Institute in 1976.
- It consists of multiple components like Base SAS, SAS/STAT, and Enterprise Miner, all specialized in data mining, predictive modeling, and statistical analysis.
- Both MATLAB and SAS are widely adopted in industry for robust machine learning implementation.

Tools and Languages in Machine Learning

Other Tools and Languages

- Several other tools support machine learning effectively. IBM SPSS, originally designed for social sciences, is now widely used across fields for statistical analysis and data mining.
- Julia, introduced in 2012, is an open-source language increasingly popular for machine learning because of its speed and ability to implement high-performance algorithms.
- Julia combines features of Python, R, and MATLAB and is gaining attention from the academic and industrial ML community.
- These additional tools complement the major ones and provide flexibility in choosing the right environment for specific domains and computational requirements in machine learning projects.

Issues in Machine Learning

Evolving Field with Diverse Challenges

- Machine learning is a relatively new and evolving field with challenges varying widely across countries due to differences in culture, legal frameworks, emotional maturity, and social norms.
- These contextual variations affect how machine learning is used and the issues that arise.
- The evolving nature of the field means ongoing adaptation is necessary to handle these diverse challenges responsibly.
- Crucially, ML use requires balancing technological advancement with respect for local laws and societal values, making its ethical deployment complex and multifaceted.

Issues in Machine Learning

Privacy and Emotional Impact

- A primary concern in machine learning is privacy. ML analyzes large amounts of historical and current data, often including sensitive personal information.
- People have different preferences for sharing such data, and inadvertent exposure or misuse might upset individuals or communities.
- Applications like targeted marketing based on ML-driven customer segmentation may emotionally hurt individuals or even lead to legal consequences in some regions.
- Thus, human judgment is essential in interpreting and using ML outcomes to avoid adverse emotional or social impacts.

Issues in Machine Learning

Need for Human Judgment and Ethical Oversight

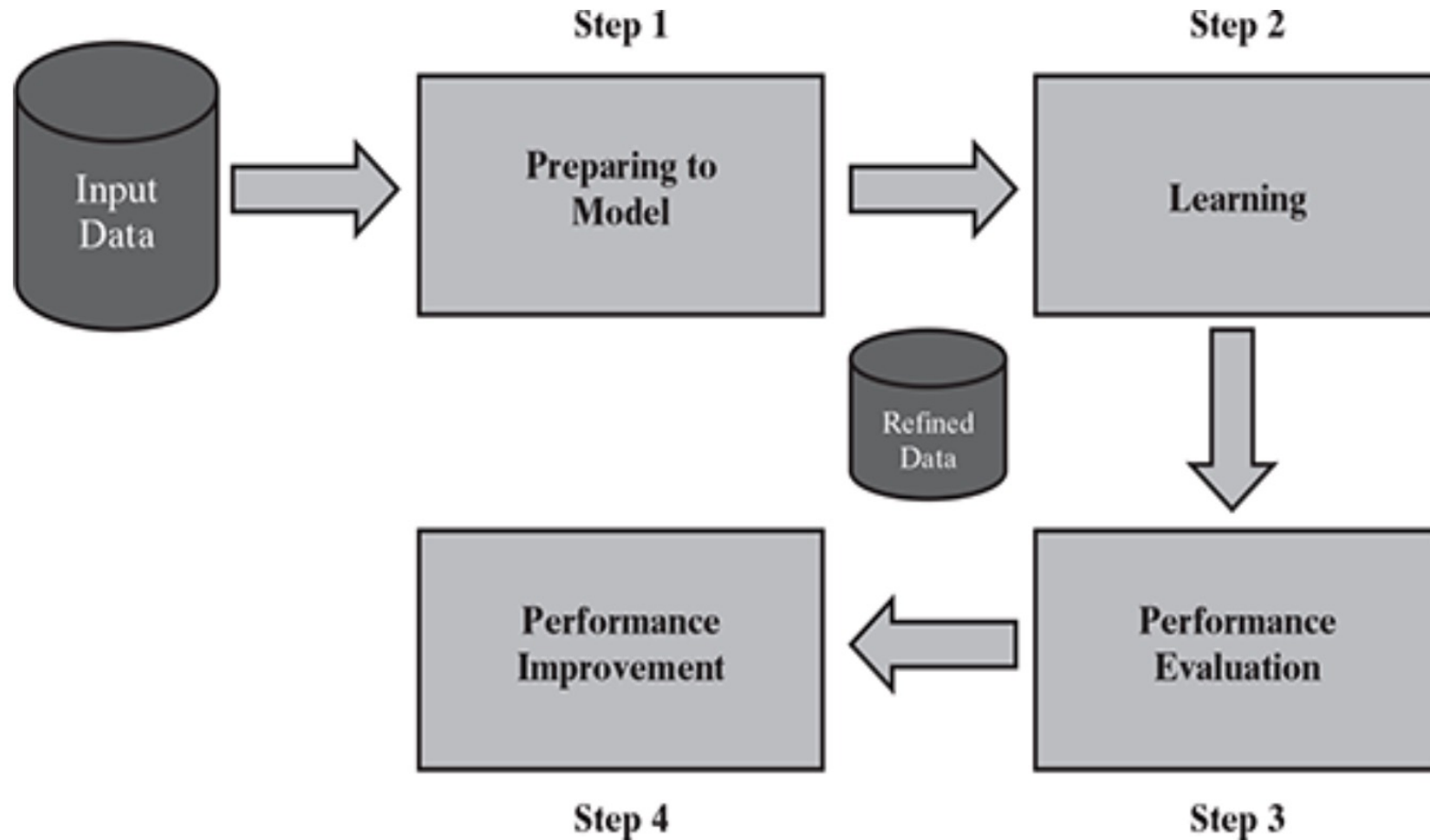
- ML predictions and decisions can lead to unintended harm if not properly overseen by humans.
- For example, political campaigns leveraging demographic insights risk alienating certain groups.
- Ethical use of ML mandates careful human evaluation before acting on ML outputs to prevent negative repercussions.
- Responsible deployment requires transparency, accountability, and consideration of fairness, bias, and social context to ensure ML benefits society without infringing rights or causing harm.
- Human judgment serves as a critical safeguard in maintaining trust and ethical standards.

Overview of Machine Learning Process

Machine learning starts with data, and the process generally involves:

1. Data preparation: Understand data type, quality, and relationships. Pre-process as needed.
2. Data split: For supervised learning, divide data into training (labelled) and test (unlabelled) sets.
3. Model selection: Choose appropriate learning algorithms based on the problem.
4. Model training & application: Train model on training data, then apply on unknown data (test data or unsupervised data).
5. Model evaluation & improvement: Assess model performance and optimize if necessary.

Overview of Machine Learning Process



Overview of Machine Learning Process

Step #	Step Name	Activities Involved
Step 1	Preparing to Model	• Understand the type of data in the given input data set • Explore the data to understand data quality • Explore the relationships amongst the data elements, e.g. inter-feature relationship • Find potential issues in data • Remediate data, if needed • Apply following pre-processing steps, as necessary: ✓ Dimensionality reduction ✓ Feature subset selection
Step 2	Learning	• Data partitioning/holdout • Model selection • Cross-validation
Step 3	Performance evaluation	• Examine the model performance, e.g. confusion matrix in case of classification • Visualize performance trade-offs using ROC curves
Step 4	Performance improvement	• Tuning the model • Ensembling • Bagging • Boosting